Do CCT Programs (Really) Reduce Mortality? Ten-Year Evidence from Mexico*

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Abstract

We study the long-term effects of Mexico's *Progresa* conditional cash transfer (CCT) program on adult mortality at ages 65 and older. Using vital statistics combined with administrative records on program enrollment, and a difference-in-differences design that exploits geographic variation in program intensity across highly marginalized municipalities, we find no evidence of medium- or long-run effects on mortality. Earlier short-run estimates suggested large mortality gains turn out to be sensitive to weighting and lag structure, becoming smaller and statistically insignificant under equally defensible specifications. To examine behavioral channels, we use *Progresa*'s own experimental evaluation data and find no significant effects on elderly labor supply or household composition.

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1 Introduction

[1] The health effects of income-support programs for older adults have attracted substantial attention, yet the evidence remains mixed. Some studies find important health benefits (Case, 2004; Barham and Rowberry, 2013; Aguila et al., 2015), others find no effects (Lloyd-Sherlock and Agrawal, 2014), and still others identify unintended adverse consequences linked to obesity and diet change (Fernald et al., 2008; Kronebusch and Damon, 2019). Most existing evidence focuses on short-run outcomes, leaving long-run mortality effects poorly understood.

[2] Mexico's *Progresa* conditional cash transfer (CCT) program, launched in 1997 and subsequently renamed Oportunidades and then Prospera, offers the opportunity to study longer-term mortality effects of a large-scale transfer program. *Progresa* has served as a model for CCT programs in over 60 countries (Baird et al., 2014; Fiszbein and Schady, 2009; Parker and Todd, 2017), and a growing literature evaluates its health impacts (Gertler, 2004; Lagarde et al., 2009; Parker and Todd, 2017; Rivera-Hernandez et al., 2016). However, long-term effects—particularly on elderly mortality—remain largely unstudied (Parker and Todd, 2017).

[3] Despite extensive evaluation of *Progresa*'s effects on children, the literature on its consequences for elderly household members is thin and almost entirely short-run. Behrman and Parker (2013) document improved health-service utilization for elderly women but not men six years after implementation. Aguila et al. (2024) find that *Progresa* improved hypertension diagnosis and treatment use among adults aged 50 and older but did not reduce uncontrolled blood pressure as measured by biomarkers, suggesting that CCTs can raise healthcare utilization without producing clinically meaningful improvements in chronic disease control. Barham and Rowberry (2013) reported short-run reductions in aggregate elderly mortality over the first five years of program operation, but no study has examined whether such effects persist or accumulate over a longer horizon. By contrast, evidence on programs that target older adults directly is more developed: Mexico's subsequent *70 y Más* non-contributory pension reduced elderly women mortality in rural areas (Menares et al., 2026) by 5.5%, and Miglino and Navarrete (2023) find that Chile's non-contributory basic pension reduced four-year mortality by 2.7 percentage points among elderly recipients. Whether a CCT structured around children's conditionalities can deliver comparable mortality benefits to the elderly over a meaningful horizon is the question we take up.

[4] These findings raise a natural question: through what channels could a CCT designed for women with young children plausibly affect elderly health? Three pathways are grounded in the program's design. First, the health conditionality requires *all* household members—including the elderly—to attend one annual clinic visit, creating an opportunity for early detection and treatment of conditions such as infectious disease, diabetes, and hypertension that disproportionately drive mortality in older age (Aguila et al., 2024; Angel et al., 2017). Second, the cash transfer raises household income by over 25% (Parker and Todd, 2017; Gertler, 2004), potentially improving nutrition and healthcare access for elderly members who share household resources. Third, if program-induced changes in labor or living arrangements alter the availability of informal caregiving within the household, this could affect elderly health through established social-support pathways (Behrman and Parker, 2013). These channels provide a credible theoretical case for expecting *Progresa* to benefit elderly household members indirectly—even though the program never targeted them directly.

[5] In this paper, we estimate difference-in-differences models exploiting variation in the proportion of households enrolled in *Progresa* across municipalities, comparing municipalities that enrolled more intensively in the program's early phase (1997–1999) to those that enrolled more intensively in the later phase (2000–2005), following the strategy of Parker and Vogl (2023). We focus on highly marginalized municipalities where the program was most heavily targeted, and study the period 1991–2006, stopping before a major structural change in 2006 when *Progresa* added a direct pension component for the elderly.

[6] Our main finding is that *Progresa* had no significant effect on elderly mortality over its first ten years of operation. This result is robust to weighting, inclusion of Seguro Popular controls (a health insurance program for informal sector workers implemented in 2003), and alternative mortality measures (levels, logs, counts, and age adjustments). We extend our analysis to cause-specific mortality, for which previous research attributed the short-run aggregate reduction to diabetes and infectious diseases, but we find no conclusive long-term significant effects, given the absence of flat pre-trends. To examine behavioral channels, we use *Progresa*'s own experimental evaluation data and find no significant effects on elderly labor supply or household composition; an attempt to analyze income and nutrition using the Mexican Household Expenditure and Income Survey similarly yields no significant results, though with limited power. As a secondary exercise, we revisit the earlier short-run evidence in Barham and Rowberry (2013) and show that the positive estimates reported for the first year's program are sensitive to weighting and lag structure: minor and similarly defensible changes to the specification attenuate them by half or more and erode statistical significance. We argue the null results are consistent with program design: *Progresa*'s benefits were primarily conditioned on children's schooling, elderly individuals living in extended households were not the direct transfer recipients, and health conditionalities required only one annual clinic visit for adults.

[7] The paper proceeds as follows. Section 2 describes the *Progresa* program. Section 3 presents the data, sample, and empirical strategy. Section 4 reports the main mortality results, the replication and cause-of-death analysis, and the mechanisms evidence. Section 5 concludes.

2 Mexico's *Progresa* CCT Program

[8] *Progresa* began in small rural communities in 1997, providing monetary transfers to poor families conditional on children attending school and family members visiting health clinics. The program rapidly expanded to cover six million families—one quarter of all Mexican families—with two thirds of beneficiaries residing in communities with fewer than 2,500 inhabitants (Parker and Todd, 2017; Fiszbein and Schady, 2009). Average monthly transfers reached approximately MXN $800 (US $90 PPP), representing more than a 25% income increase for many poor households (Gertler, 2004; Parker and Todd, 2017). All transfers are paid to the female head of household.

[9] The program includes two main grant types: grants conditional on children's school attendance (for children under 22), and a health and nutrition transfer requiring all family members to attend regular health clinic visits. For adults aged 17 and older, including the elderly, the attendance requirement is one clinic visit per year—a minimal conditionality. In 2006, a pension component for adults aged 70 and older was added, fundamentally altering the program's reach for the elderly. We end our analysis in 2006 to focus on the original CCT design.

[10] The majority of elderly individuals in rural eligible households live in extended families (Behrman and Parker, 2013). Before the program, approximately 75% of elderly individuals lived in households with children under 15 (Behrman and Parker, 2013). In most cases, elderly individuals are grandparents of children receiving educational grants and thus do not directly control program expenditures, though they may benefit indirectly. Given this structure, large and immediate effects on elderly health and mortality would not be expected.

[11] Elderly mortality in Mexico has declined over time and is slightly higher in non-marginalized areas than in marginalized (poor) areas, as shown in Figure 1, and Appendix Figure A.1. Beltrán-Sánchez and Wong (2025) demonstrate using panel data from the Mexican Health and Aging Survey that this pattern is not explained by selective migration of unhealthy individuals toward urban areas.

3 Data and Empirical Strategy

3.1 Data

[12] Mortality data. We use individual-level death registries derived from death certificates, covering every death in Mexico between 1991 and 2006—over three million records. The data include cause of death, birth year, sex, and the municipality of residence of the deceased, which we use to link to population denominators and program variables. All data are publicly available from the Mexican National Institute of Statistics and Geography (INEGI).[1]

[13] The expansion of Seguro Popular—Mexico's public health insurance program for informal-sector workers, which rolled out staggered across municipalities beginning in 2002—may have improved the accuracy of death registration. To address this, we control for Seguro Popular enrollment intensity at the municipality level.[2]

[14] Population data. We use INEGI population counts disaggregated by sex, municipality, and age from Census data (1990, 2000, 2005, 2010, 2020), linearly interpolating annual estimates for each sex-age-municipality cell. We similarly interpolate municipal household counts to construct program intensity denominators.

[15] *Progresa* administrative data. We obtain annual counts of enrolled *Progresa* households by municipality from administrative records, which we use to construct our treatment intensity variable.

[16] Additional controls. We use the 1990 CONAPO Municipal Marginalization Index to restrict the sample to highly marginalized areas.[3] We also incorporate administrative data on Seguro Popular beneficiaries to construct a municipality-year enrollment intensity variable.

[17] *Progresa* experimental survey data. To examine short-run mechanisms, we use the *Progresa* evaluation surveys—the baseline household survey (ENCASEH, 1997) and follow-up rounds

[1] INEGI derives mortality data from the certification system of the Mexican Ministry of Public Health. Mexico generally maintains high-quality vital statistics (Mahapatra et al., 2007; Mikkelsen et al., 2015).

[2] Seguro Popular reached full municipal coverage by 2006 but significant portions of the eligible population remained uninsured throughout our study period.

[3] The Marginalization Index classifies all Mexican municipalities into five categories based on household poverty indicators, including dirt floors, crowding, illiteracy, and lack of sanitation infrastructure.

(ENCEL, 1998–2000)—collected in 506 treatment and 320 control localities across seven Mexican states (Guerrero, Hidalgo, Michoacán, Puebla, Querétaro, San Luis Potosí, and Veracruz). Treatment localities began receiving *Progresa* transfers in early 1998; control localities were enrolled in late 1999, providing an 18-month experimental window for comparison. The surveys collected individual-level information on labor supply, household composition, and income for all household members. Our mechanisms sample is restricted to elderly individuals aged 65 and older living in eligible households, identified using the program's eligibility criteria applied to the 1997 baseline.

3.2 Sample and Descriptive Statistics

[18] Our analysis sample is restricted to deaths among adults aged 65 and older, in highly marginalized municipalities classified as high (grade 4) or very high (grade 5) on the 1990 CONAPO Marginalization Index. We construct municipality-year cells combining death counts with population denominators and program variables. Table 1 reports summary statistics comparing marginalized and non-marginalized municipalities in the pre-program period. As expected, marginalized municipalities have substantially worse socioeconomic conditions and much higher *Progresa* enrollment rates.

3.3 Empirical Strategy

[19] We exploit variation in the proportion of households enrolled in *Progresa* across municipalities before and after the program began in 1997.[4] Following Parker and Vogl (2023), who study the long-run effects of *Progresa*, we distinguish between municipalities that enrolled households more intensively in the early phase (1997–1999, denoted $Intensity_{1999}$) and those that enrolled more intensively in the later phase (2000–2005, denoted $Intensity_{2005}$). They argue that after 2005 program expansion shifted substantially toward urban areas, making the two phases distinct and largely non-overlapping sources of intensity variation across municipalities.[5]

We adopt the same approach and apply it to elderly mortality in using municipality-level intensity because (i) the municipality is the finest geographic unit at which mortality data and population

[4] *Progresa* enrollment was originally assigned at the locality level, following a targeting algorithm based on community-level poverty scores (Parker and Todd, 2017). Beneficiary counts are, however, available and used at the municipality level, which aggregates multiple localities.

[5] Appendix Figures A.1 and A.2 document mortality trends and the spatial distribution of *Progresa* intensity across all municipalities and separately for the highly marginalized subsample, confirming substantial variation in both early-phase (1999) and late-phase (2005) enrollment.

denominators are consistently observed across our 1991–2006 panel; (ii) aggregating intensity to the municipality level averages over within-municipality variation in program rollout, reducing noise from locality-level targeting decisions; and (iii) municipal coverage rates better capture population-level exposure.[6]

[20] Because *Progresa* was preferentially targeted toward more disadvantaged areas, a simple difference-in-differences comparing *Progresa* municipalities to all others could be confounded by differential trends. Restricting the sample to highly marginalized municipalities mitigates this concern. Our identification strategy further refines variation to capture the impact of earlier versus later cumulative exposure to the program (rather than ever-versus-never), under the assumption that mortality effects are cumulative. We estimate:

$$\text{MR}_{m,t} = \beta_0 \, Post_t \times Intensity_{m,1999} + \beta_1 \, Post_t \times Intensity_{m,2005} + \delta_m + \gamma_t + \chi_{m,t} + \varepsilon_{m,t} \qquad (1)$$

where $\text{MR}_{m,t}$ is the mortality rate (deaths per 1,000 inhabitants aged 65+) in municipality m in year t. $Post_t$ equals one for years 1997–2006 and zero for 1991–1996. Municipality fixed effects δ_m control for time-invariant unobservables; year fixed effects γ_t absorb common time trends. $\chi_{m,t}$ is the Seguro Popular enrollment intensity, included in specifications that control for this potentially confounding program. Standard errors are clustered at the municipality level. Throughout this analysis, our coefficient of interest is β_0, which captures the effect of early-phase enrollment intensity ($Intensity_{1999} \times Post$); β_1 controls for later-phase enrollment, as previously described.[7] We retain the level outcome in the main analysis for three reasons: it eases interpretation (effects in deaths per 1,000), it preserves comparability across outcome definitions and with prior work, and it avoids discarding municipality-year cells with zero deaths that a log transformation would require.[8]

[21] We also estimate event study specifications to trace dynamic effects and test for differen-

[6] Staggered adoption estimators (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; d'Haultfoeuille, d'Haultfoeuille) require a clearly defined binary treatment date for each unit. Our setting does not satisfy this requirement: *Progresa* was rolled out gradually within municipalities, and the intensity measure captures cumulative enrollment rather than a discrete entry date.

[7] The mortality rate equals zero in a non-trivial share of municipality-year cells, particularly in small municipalities with few elderly residents. We address this in the appendix by estimating the same specification using raw death counts (Poisson regression) and log-transformed outcomes—both yield qualitatively similar results to the level specification.

[8] Chen and Roth (2024) formalize the problems that arise when taking logs in the presence of zeros and recommend count-based alternatives; while Poisson regression addresses these concerns, we report it only as a robustness check to maintain comparability with prior literature.

tial pre-trends:

$$\text{MR}_{m,t} = \sum_{k \neq 1996} \beta_k \mathbf{1}[t = k] \times Intensity_{m,1999} + \sum_{k \neq 1996} \theta_k \mathbf{1}[t = k] \times Intensity_{m,2005} + \delta_m + \gamma_t + \chi_{m,t} + \varepsilon_{m,t} \quad (2)$$

with 1996 as the reference year. The β_k coefficients trace the effect of early-phase enrollment intensity on mortality in each year relative to 1996.

[22] Our interaction design, following Parker and Vogl (2023), anchors identification to a discrete break in 1997 and explicitly controls for later-phase enrollment, separating the effects of the two rollout phases, and allowing us to include all highly marginalized municipalities without a phase-in restriction. The event study format of equation (2) additionally allows us to test for differential pre-trends—a test. We include also include population weights because *Progresa* targeted the most isolated and impoverished communities first, unweighted estimates may overstate the average effect by giving excess weight to the highest-intensity, smallest municipalities, whose mortality rates are also most volatile. Weighting by the elderly population gives each municipality influence proportional to its contribution to total elderly deaths.

[23] Our difference-in-differences coefficient β_0 in equation (1) measures the average effect of a one-unit increase in early-phase enrollment intensity on the mortality rate for adults aged 65 and older, and aligns with the population-level policy question: what was the average effect of *Progresa* on the mortality of elderly Mexicans living in marginalized areas? Identification relies on the assumption that, conditional on municipality and year fixed effects, municipalities with higher $Intensity_{1999}$ would have followed parallel mortality trends to those with lower intensity in the absence of the program.[9]

[9]Restricting to highly marginalized municipalities means the coefficient estimates an Intent-to-Treat (ITT) effect: the reduced-form impact of municipal-level program exposure on elderly mortality, regardless of whether individual elderly residents' households directly received transfers. Since Progresa primarily targeted families with school-age children, any effect on elderly mortality operates through indirect channels — household income spillovers, improved local health access, or caregiver effects. The sample restriction ensures this ITT is estimated within Progresa's intended geographic target population, making it policy-relevant for assessing the program's reach.

4 Results

4.1 Main Results

[24] Table 2 reports difference-in-differences estimates of the effect of *Progresa* enrollment intensity on the mortality rate for adults aged 65 and older. The bottom rows of the table report the mean dependent variable and mean intensity values, providing the scale needed to interpret the coefficients. As documented in Figure 1 and Figure 1, enrollment intensity varied substantially across municipalities—from near zero in the pre-program period to meaningful coverage by 1999 and further by 2005—with the geographic spread and temporal dynamics that motivate the two-phase design. Panel A reports pooled results, Panel B results for females, and Panel C results for males. Within each panel, column (1) presents unweighted estimates without Seguro Popular controls; column (2) adds Seguro Popular intensity; column (3) presents population-weighted estimates (weighted by the elderly population); and column (4) combines weighting with Seguro Popular controls.

[25] Across all specifications, columns, and panels, the estimated effect of *Progresa* enrollment intensity on elderly mortality is small and statistically insignificant. The interaction of early-phase intensity ($Intensity_{1999} \times Post$) is close to zero in both the unweighted and weighted specifications, for the pooled sample and separately by sex. Adding Seguro Popular controls in columns (2) and (4) does not materially alter results. These findings hold regardless of whether we weight by the size of the elderly population.

[26] Figure 2 presents the event study estimates for the weighted specification with Seguro Popular controls (column 4). The pre-program coefficients do not show a clear and stable point estimate close to zero, and thus statistically significant differential trends, which prevent us from supporting the parallel trends assumption underlying our design. In the post-program period, while some individual year-specific coefficients—particularly those associated with males following the 1997 rollout—are negative and individually significant, we do not have enough evidence to conclude that the possible transitory decline in mortality in those years is attributable to the program, consistent with the null average DiD estimate documented in Table 2. For females, while it seems to show no significant pre-trends, post-period effects are mostly zero.

[27] **Cause-of-death analysis** Following previous work suggesting cause of death paths on mortality reduction, we decompose the aggregate null result into nine cause-specific components:

cancer, cardiovascular disease, respiratory disease, infectious disease, diabetes, nutritional causes, ill-defined causes, accidents, and other. Appendix Table A.1 and Figures A.3–A.4 present DiD estimates and event studies for this exploratory decomposition. Early-phase enrollment ($Intensity_{1999} \times Post$) yields negative point estimates for cancer (pooled: -0.887, $p < 0.01$), ill-defined causes (-0.574, $p < 0.01$), and diabetes (-0.508, $p < 0.05$), alongside a positive point estimate for cardiovascular mortality ($+1.530$, $p < 0.10$). However, none of these results survives Romano-Wolf step-down correction for the familywise error rate across the nine hypotheses tested jointly (all adjusted p-values > 0.05, except for cancer), indicating that the individually significant estimates likely reflect multiple-testing variation rather than genuine cause-specific impacts. The significant coefficient on ill-defined causes warrants additional caution, as it may capture changes in cause-of-death coding or registration practices over the panel rather than true mortality reductions. The event studies in Appendix Figures A.3–A.4 are broadly consistent with this interpretation: post-period confidence intervals overlap substantially with the pre-period distribution for most causes, providing no clear visual evidence of a break at program rollout. We therefore conclude that the cause-specific results are consistent with the aggregate null rather than indicative of heterogeneous program effects that cancel in the aggregate.

4.2 Robustness

[28] We first assess robustness to functional form and outcome definition. Appendix Table A.2 presents DiD estimates under four specifications: column (1) is the baseline mortality rate in levels (weighted); column (2) uses the natural log of the mortality rate; column (3) uses Poisson pseudo-maximum likelihood with raw death counts; and column (4) replaces the mortality rate with the Age-Adjusted Mortality Rate (AAMR), which directly standardizes for the age structure of each municipality's elderly population. All regressions are weighted and include Seguro Popular intensity. Estimates remain small and statistically insignificant across all columns, mirroring the structure of Table 2. Appendix Figure A.5 presents the corresponding event study estimate; the pre- and post-period patterns are qualitatively similar to the baseline level specification in all cases.

[29] We next examine the relationship between our long-run findings and the short-run results of Barham and Rowberry (2013) (hereafter BR). BR provided the first systematic evidence that *Progresa* reduced elderly mortality, using a design in which mortality is regressed on a two-year lagged enrollment intensity measure without a pre/post interaction:

$$\text{MR}_{m,t} = \alpha_0 + \alpha_1\, Intensity_{m,t-2} + \delta_m + \gamma_t + \chi_{m,t} + \varepsilon_{m,t} \tag{3}$$

[30] Appendix Table A.3 reproduces their exercise, restricting to municipalities incorporated into *Progresa* in 1998 or 1999. Row (A) reports their original pooled estimate; row (B) shows our close replication with a comparable coefficient. This confirms that data and procedural differences are not the source of the divergence between their results and ours. When population weights are added in row (C), the estimate attenuates by approximately 50% and loses precision, reflecting the disproportionate influence of small, high-intensity municipalities in the unweighted specification. Rows (D) and (E) show that replacing the two-year lag with one- or three-year alternatives further reduces the estimated coefficient, underscoring the sensitivity of the finding to this specification choice.

[31] To connect their approach to ours, and thus be able to test whether there are pretrends, we apply the interaction design of equation (1) to the same BR sample and window, replacing the time-varying lagged intensity with a fixed early-phase measure interacted with a post-1997 dummy:

$$\text{MR}_{m,t} = \beta_0\, Post_t \times Intensity_{m,1999} + \delta_m + \gamma_t + \varepsilon_{m,t} \tag{4}$$

[32] Appendix Table A.4 reports estimates under this design for both the BR sample and the full highly marginalized sample over 1992–2002. In the BR sample, the unweighted estimate is negative and significant, but adding population weights attenuates it substantially—consistent with the sensitivity documented in Appendix Table A.3. In the highly marginalized sample, the effect is no longer statistically significant in any specification. Appendix Figure A.6 directly compares event study estimates across both sample definitions and by sex, illustrating how the sample restriction and weighting choices jointly account for the divergence between the short-run finding and our long-run null.

[33] Cause-specific results for the Barham-Rowberry sample over 1992–2002 are presented in Appendix Table A.5 and Figures A.7–A.8. Early-phase enrollment significantly reduces infectious disease (-1.670, $p < 0.01$), diabetes (-0.924, $p < 0.01$), cancer (-0.816, $p < 0.01$), and ill-defined causes (-0.614, $p < 0.01$) in the BR sample. This extends the interpretation of Barham and Rowberry (2013)—who attributed their aggregate mortality reduction primarily to infectious disease and diabetes—by showing that cancer and ill-defined mortality were also declining in the early program years. Cardiovascular mortality is positive and marginally significant (+0.792, $p < 0.10$), as in the main sample. Across both samples, cause-specific patterns point to reductions in conditions amenable to preventive and primary care, partially offset by cardiovascular increases.

[34] To put the magnitudes in perspective, BR report a gain of approximately one year of life expectancy at age 65 from their baseline specification. Our close replication yields a similar negative coefficient in the unweighted specification; adding population weights halves the estimate. Using a constant-hazard approximation against a pre-program mean mortality rate of approximately 55 per 1,000 (Table 2), our weighted replication corresponds to a gain of a few months of life expectancy at most, with a confidence interval that includes zero.[10] Our long-run design, covering all highly marginalized municipalities through 2006, finds no statistically significant effect, indicating that any transitory short-run gain did not persist over the longer horizon.

4.3 Mechanisms: Labor Supply, Household Composition, and Income Effects

[35] We take additional steps to examine the behavioral channels through which *Progresa* could plausibly have affected elderly health. The theoretical case for expecting an effect rests on three program-specific pathways. The most direct is the *health conditionality*: *Progresa* required all household members, including the elderly, to attend one annual clinic visit, which could enable early detection of infectious disease, diabetes, and hypertension—precisely the causes BR identify as driving their short-run mortality reduction. Behrman and Parker (2013) document that this conditionality increased health-service utilization among elderly women, and **?** confirm that it improved hypertension diagnosis and treatment uptake, supporting the hypothesis that the clinical contact pathway was active. The second pathway is *income and nutrition*: a 25%-plus increase in household income could reduce nutritional deficiencies and improve access to medications and outpatient care for elderly members who share household resources. The third pathway operates through *living arrangements*: if *Progresa*'s transfers changed co-residence patterns—for instance, by reducing the need for adult children to migrate for work—elderly household members could benefit from increased informal caregiving and social support, both of which are associated with

[10] Using a constant-hazard (exponential) survival approximation: $e_{65} = 1/\mu$ where μ is the annual mortality rate. Against a pre-program mean mortality rate of approximately 55 per 1,000, the baseline life expectancy is $1/0.055 \approx 18.2$ years. BR's reported one-year gain implies a mortality reduction of approximately 3 per 1,000. Our weighted replication's smaller coefficient implies a correspondingly smaller gain. These calculations involve different samples and specifications, so the comparison should be interpreted with that caveat in mind.

lower elderly mortality (Duflo, 2000). We test the second and third channels directly; the first is addressed by the evidence in **?** and the cause-of-death analysis above.

[36] Table 3 uses experimental data from eligible households in the original 1997 *Progresa* rollout to examine the short-run behavioral responses in labor supply and household composition channels. Column (1) reports the effect on weekly hours worked by elderly individuals; columns (2)–(4) report effects on the probability of living alone, living with children, and living exclusively with other elderly members, respectively. The estimates show no significant changes: *Progresa* did not reduce elderly labor hours and did not systematically alter whether elderly lived with children or alone. The null on living arrangements is informative—it rules out the informal-caregiving pathway as an operative mechanism, at least within the experimental sample and time horizon. Appendix Table **??** extends this analysis to household expenditure outcomes stratified by elderly household presence; the estimated differential effects for households with elderly members are similarly imprecise and not statistically significant.

[37] Complementing the experimental evidence, we examine the income and nutrition channels using the National Household Income and Expenditure Survey (ENIGH), drawing on the mechanisms framework of Menares et al. (2026) for the *70 y Más* pension program. We estimate the effects of *Progresa* intensity on elderly households' labor income, food consumption by category, and out-of-pocket health spending.[11] All estimated coefficients are imprecise with large standard errors that preclude meaningful inference. These null ENIGH results do not contradict the experimental null from Table 3—they reinforce it, providing convergent evidence from two independent data sources that the income, nutrition, and caregiving channels through which *Progresa* might have reached elderly members were not activated in ways large enough to reduce mortality.

5 Conclusion

[38] This paper re-examines the medium-term effect of Mexico's *Progresa* conditional cash transfer program on elderly mortality. Using fifteen years of vital statistics and administrative enrollment records, we find no evidence that *Progresa* significantly reduced mortality among adults aged 65 and older during its first ten years of operation.

[11] We estimate difference-in-differences models using ENIGH waves from 1992 to 2006, exploiting variation in *Progresa* enrollment intensity across municipalities, following the same two-phase interaction design as in equation (1). Outcome variables include individual labor income, household food expenditure by category (cereals, meat and dairy, vegetables, sugar and fats), and health spending (outpatient visits, medications, inpatient care).

[39] These null results emerge coherently from the evidence on mechanisms. As described in Section 4.3, a CCT targeting women with children can plausibly reach the elderly through three channels: the annual health checkup conditionality, household income and nutrition effects, and changes in co-residence and informal care. The evidence suggests the first channel was partially activated—Behrman and Parker (2013) document improved health utilization for elderly women and **?** confirm higher hypertension diagnosis and treatment uptake—yet these gains did not translate into improved biomarkers or, as we show, lower mortality. The second and third channels show no activation at all: Table 3 finds no significant effects on elderly labor supply or living arrangements, and our ENIGH-based income and nutrition analysis is too imprecise to detect behavioral responses, likely because the survey was not designed to capture elderly subgroups in small rural municipalities. Taken together, the picture is of a program that nudges elderly household members toward the healthcare system once a year but does not produce the sustained changes in income, nutrition, or caregiving that would be needed to reduce mortality over a decade. The health checkup conditionality—one annual visit—is simply too minimal a clinical intervention to drive the mortality reductions that a pension transferring resources directly to older adults can achieve (Menares et al., 2026).

[40] We further show that the key prior finding of Barham and Rowberry (2013) is sensitive to specification. Their result attenuates by 50% with population weighting and shrinks further with alternative lag structures, losing statistical significance. Combined with our own estimates, we conclude that the evidence for large mortality effects of conditional cash transfers on older adults is not strong. In contrast, Menares et al. (2026) find that Mexico's *70 y Más* non-contributory pension—which transferred resources directly and unconditionally to adults aged 70 and older—did reduce elderly women elderly mortality in rural areas by 5.5%, and Miglino and Navarrete (2023) document a 2.7 percentage-point reduction in four-year mortality from Chile's non-contributory basic solidarity pension for the elderly. These contrasting findings point to a fundamental difference in how program benefits reach older adults.

[41] When income arrives via a child-focused CCT, any gain to the elderly depends on the household's willingness and ability to redistribute resources across generations—resources that are in the first instance conditioned on children attending school. When income arrives as a direct, unconditional pension to the older adult, none of that intermediation is needed. The evidence suggests that households receiving *Progresa* did not systematically redirect income toward elderly members' food, healthcare, or living arrangements, at least not enough to produce detectably lower mortality.

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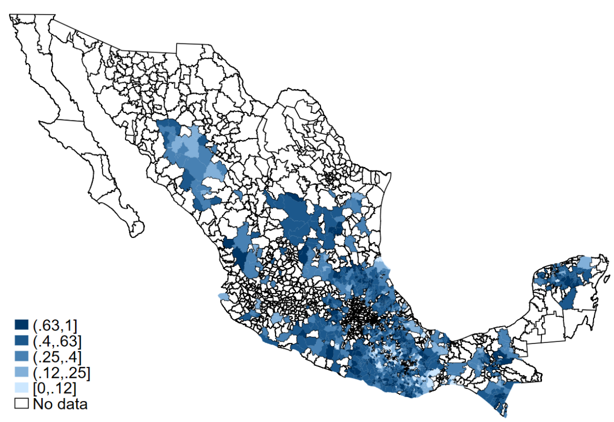
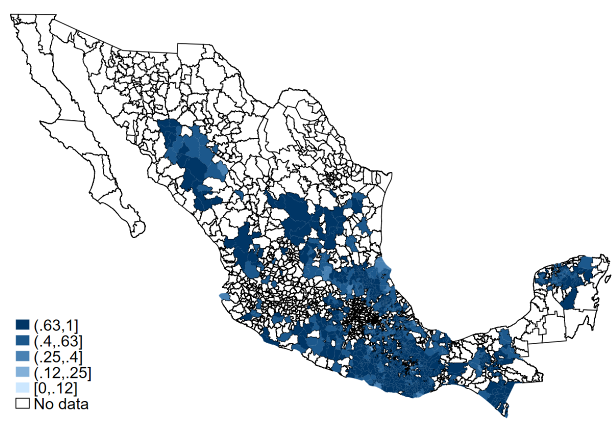
Figure 1: Progresa Enrollment Intensity and Mortality Rate Trends

(a) Mortality and Progresa Intensity



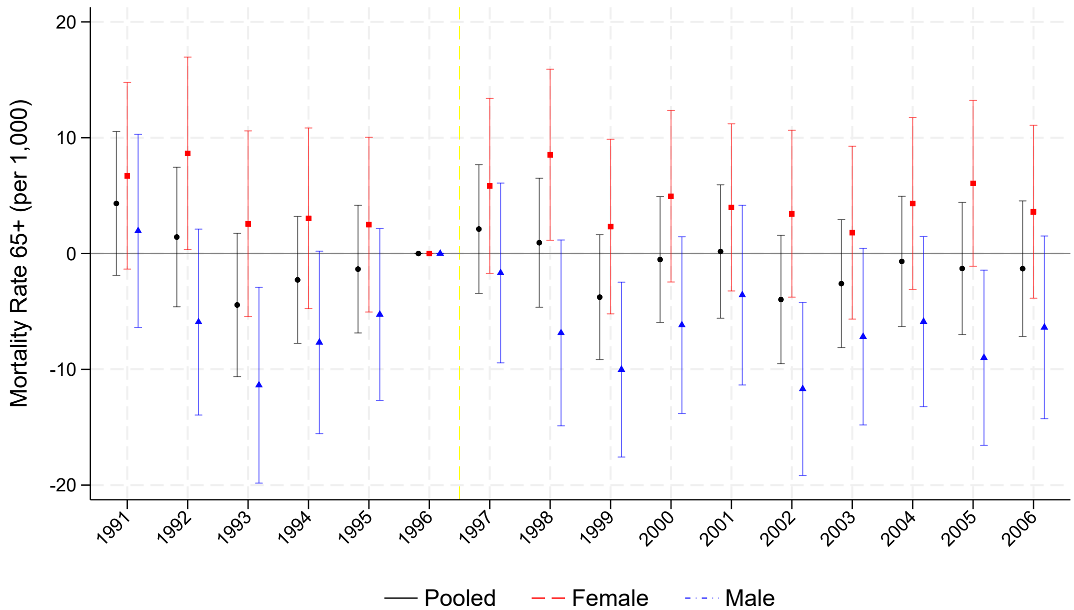
(b) Intensity in 1999



(c) Intensity in 2005

Notes: Panel (a) plots the mortality rate for adults aged 65 and older—shown separately for all, male, and female—alongside Progresa penetration for highly marginalized municipalities over 1991–2006. The vertical dashed line marks 1997, the year of initial program rollout. The left axis measures deaths per 1,000 population aged 65 and older; the right axis measures Progresa penetration as the share of beneficiary households in total municipal households. Panels (b) and (c) display municipality-level Progresa intensity for 1999 and 2005, respectively, among highly marginalized municipalities. Darker shading indicates higher program intensity. The sample is restricted to municipalities classified as high or very high on the 1990 CONAPO Marginalization Index.

Figure 2: Event Study: Effect of Progresa Enrollment Intensity on Mortality Rate by Sex



(a) Weighted

Notes: This figure shows the results obtained from estimating a difference-in-differences event study specification. The outcome is the Mortality Rate for adults aged 65 and older, measured in deaths per 1,000 inhabitants. The plots display the interaction coefficients between Progresa intensity in 1999 and year indicators. The specification includes municipality and year fixed effects and controls for Seguro Popular intensity. Standard errors are clustered at the municipality level. The reference year is 1996. The sample is restricted to highly marginalized municipalities (*gm_mun_1990* = 4 or 5). Regressions are weighted by the population aged 65 and older. 95% confidence intervals shown.

Table 1: Summary Statistics: Marginalized and Non-Marginalized Municipalities in Mexico, Pre-Program Period

	Non-marginalized areas	**Marginalized areas**
A. Progresa program intensity		
1999	0.10 (0.13)	0.44 (0.22)
2005	0.28 (0.19)	0.72 (0.17)
B. Characteristics of municipalities, 1990		
Components of the marginality index		
Illiterate population	13.33 (6.57)	33.33 (13.3)
With less than primary education	45.74 (12.32)	69.56 (9.68)
Without a toilet	25.94 (16.62)	60.30 (18.54)
Without electricity	11.81 (10.29)	36.39 (24.72)
Without running water	19.43 (14.87)	50.65 (24.07)
With crowding	59.84 (9.79)	73.96 (8.24)
With dirt floor	22.05 (14.21)	61.98 (21.58)
In communities with $<5{,}000$ inhabitants	60.18 (36.15)	95.50 (13.54)
Earning $<2\times$ minimum wage	69.19 (11.60)	85.82 (8.55)
Number of municipalities	1,234	1,119

Notes: Panel A reports the proportion (%) of municipalities with Progresa beneficiaries, defined as the ratio of beneficiary households to total municipal households, with standard deviations in parentheses. Panel B presents sample means (%) for socioeconomic characteristics with standard deviations in parentheses. Marginalized municipalities are those classified as high (grade 4) or very high (grade 5) on the 1990 CONAPO Marginalization Index; non-marginalized municipalities comprise those with medium, low, or very low marginalization levels. "With crowding" is measured as the number of occupants per room.

	Marginalized	Non-Marginalized
	(1)	(2)
Panel A: Progresa Enrollment Intensity (%)		
Intensity in 1999	44.4	10.3
	(21.9)	(12.7)
Intensity in 2005	71.7	28.5
	(16.7)	(19.4)
Panel B: Socioeconomic Characteristics (%)		
% illiterate	29.0	11.5
	(11.9)	(5.9)
% without completed primary	62.4	39.9
	(9.7)	(11.7)
% without electricity	20.2	5.8
	(17.8)	(6.3)
% without piped water	35.6	12.4
	(23.2)	(11.9)
With crowding	68.7	52.4
	(9.4)	(10.1)
% with dirt floors	54.4	17.7
	(22.2)	(12.5)
% without drainage	67.6	30.6
	(16.8)	(19.8)
% in localities <5,000	94.2	58.1
	(14.8)	(35.7)
Municipalities	1119	1234

Notes: Panel A reports the proportion (%) of municipalities with Progresa beneficiaries, defined as the ratio of beneficiary households to total municipal households, with standard deviations in parentheses. Panel B presents sample means (%) for socioeconomic characteristics with standard deviations in parentheses. Marginalized municipalities are those classified as high (grade 4) or very high (grade 5) on the 1990 CONAPO Marginalization Index; non-marginalized municipalities comprise those with medium, low, or very low marginalization levels. "With crowding" is measured as the number of occupants per room.

Table 2: Effects of Progresa Enrollment Intensity on Mortality Rate by Sex

	(1)	(2)	(3)	(4)
Panel A: Pooled				
Intensity 1999 x post (1997-2006)	0.369	0.354	-0.777	-0.719
	(2.546)	(2.547)	(1.464)	(1.464)
Intensity 2005 x post (1997-2006)	-5.525*	-5.513*	-2.787	-2.796
	(3.075)	(3.074)	(1.806)	(1.803)
Mean (1991-1996)	46.47	46.47	46.47	46.47
Obs	17,904	17,904	17,904	17,904
Panel B: Females				
Intensity 1999 x post (1997-2006)	1.737	1.668	0.592	0.620
	(2.932)	(2.928)	(1.824)	(1.824)
Intensity 2005 x post (1997-2006)	-9.117**	-9.063**	-5.637**	-5.641**
	(3.776)	(3.770)	(2.205)	(2.205)
Mean (1991-1996)	46.11	46.11	46.11	46.11
Obs	17,904	17,904	17,904	17,904
Panel C: Males				
Intensity 1999 x post (1997-2006)	-1.627	-1.588	-2.288	-2.206
	(2.767)	(2.776)	(1.732)	(1.735)
Intensity 2005 x post (1997-2006)	-1.488	-1.519	0.214	0.201
	(3.309)	(3.312)	(2.000)	(1.994)
Mean (1991-1996)	47.26	47.26	47.26	47.26
Obs	17,904	17,904	17,904	17,904
No. Mun	1,119	1,119	1,119	1,119
Seguro Popular	N	Y	N	Y
Weights	N	N	Y	Y
Mean Intensity 1999 (%)	44.4	44.4	44.4	44.4

Notes: This table reports difference-in-differences estimates of the effect of Progresa enrollment intensity on the Mortality Rate for adults aged 65 and older per 1,000 inhabitants. Columns (1) and (2) report unweighted estimates without and with Seguro Popular intensity, respectively; columns (3) and (4) report estimates weighted by the population aged 65 and older without and with Seguro Popular intensity. $Intensity_{1999} \times Post$ captures the differential effect of early-phase (1997–1999) Progresa enrollment: a coefficient of one corresponds to moving from 0% to 100% beneficiary household coverage after 1997. $Intensity_{2005} \times Post$ captures the analogous effect for the later phase (2000–2005). The post-dummy equals 1 for years 1997–2006. The sample is restricted to highly marginalized municipalities (*gm_mun_1990* = 4 or 5). All regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level and shown in parentheses. ***p< 0.01, **p< 0.05, *p< 0.1.

Table 3: Effects of Progresa on Labor Supply and Living Arrangements of Elderly in Eligible Households

	(1) Weekly Hours	(2) Live Alone	(3) With Children	(4) Only Elderly
Panel A: Females				
Treat × 1998	0.124	0.000	0.001	−0.008
	(1.136)	(0.008)	(0.011)	(0.008)
Treat × 1999	0.292	0.007	0.015	0.012
	(1.070)	(0.010)	(0.018)	(0.014)
Observations	3,338	3,597	3,597	3,597
Control Mean (1997)	3.923	0.044	0.708	0.145
Panel B: Males				
Treat × 1998	−1.210	0.012	−0.008	−0.007
	(1.895)	(0.008)	(0.011)	(0.010)
Treat × 1999	0.378	0.008	0.040**	−0.001
	(1.872)	(0.012)	(0.020)	(0.017)
Observations	3,170	3,419	3,419	3,419
Control Mean (1997)	25.288	0.049	0.676	0.159

Notes: This table reports difference-in-differences estimates of the short-run effect of Progresa on labor supply and living arrangements for elderly individuals aged 65 and older in eligible households. The sample is restricted to households eligible for Progresa. Column (1) reports weekly hours worked. Columns (2)–(4) report living arrangement outcomes: living alone, living with children, and living only with other elderly members, respectively. The baseline year is 1997. Standard errors are clustered at the municipality level and shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Online Appendix

Do CCT Programs (Really) Reduce Mortality: Ten-Year Evidence from Mexico

Emma Aguila, William H. Dow, Felipe Menares, Susan Parker, Jorge Peniche, and Soomin Ryu

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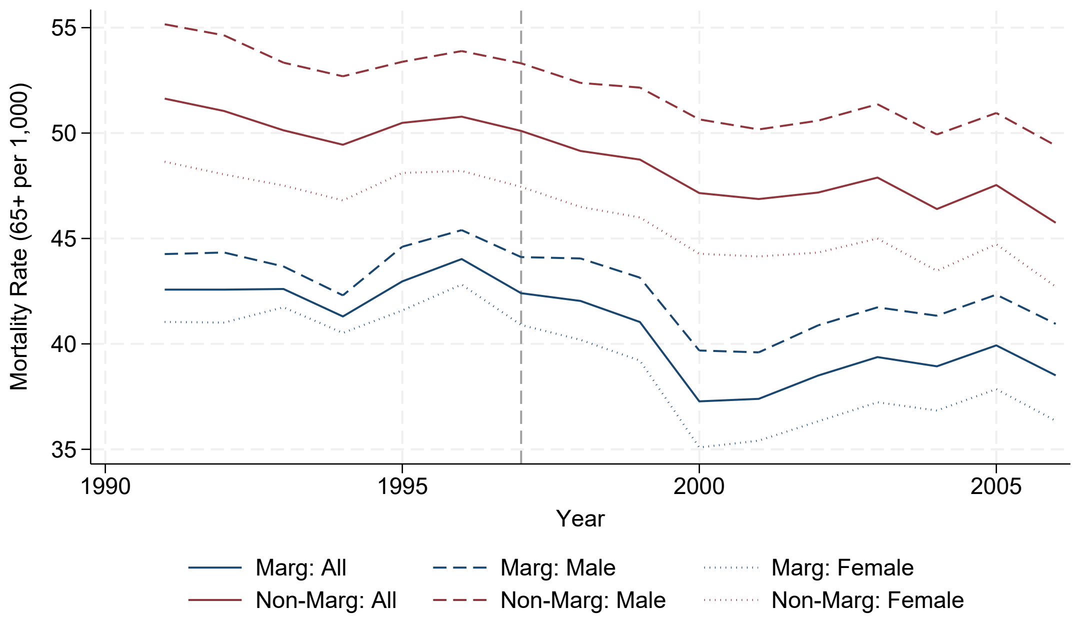
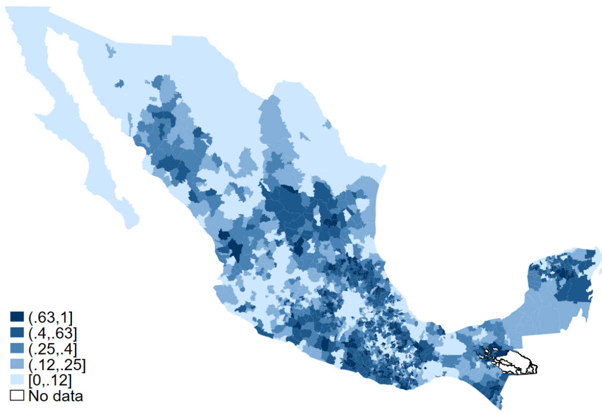
Appendix A: Additional Figures and Tables

Figure A.1: Progresa Enrollment Intensity and Mortality Rate Trends

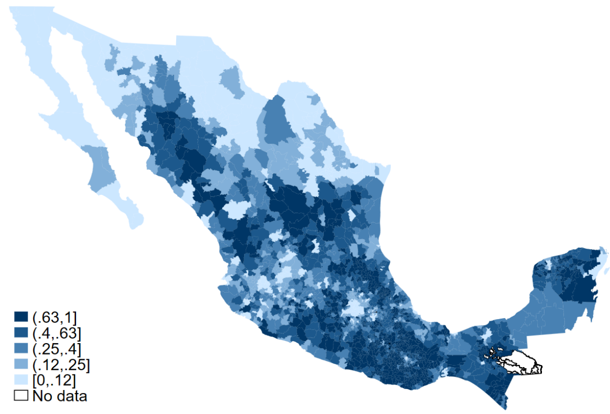


Notes: This figure documents trends variation in Progresa enrollment for all municipalities, complementing Figure 1 in the main text which focuses on the highly marginalized subsample. Panel (a) plots the mortality rate for adults aged 65 and older alongside Progresa penetration across all Mexican municipalities over 1991–2006. Comparing panel (a) here with Figure 1(a) shows that mortality trends are broadly similar across the two samples, supporting the external relevance of the main analysis.

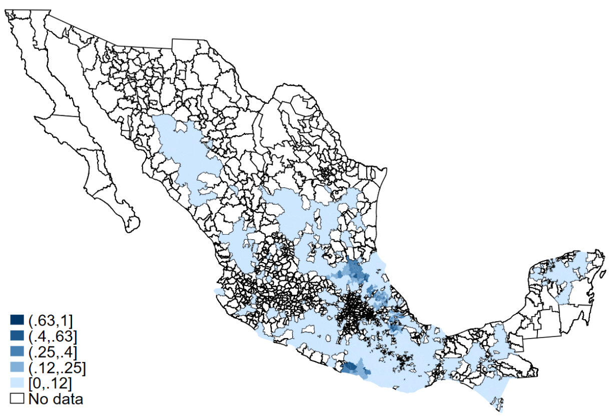
Figure A.2: Progresa Enrollment Intensity and Mortality Rate Trends



(a) Intensity in 1999 - All Municipalities



(b) Intensity in 2005 - All Municipalities

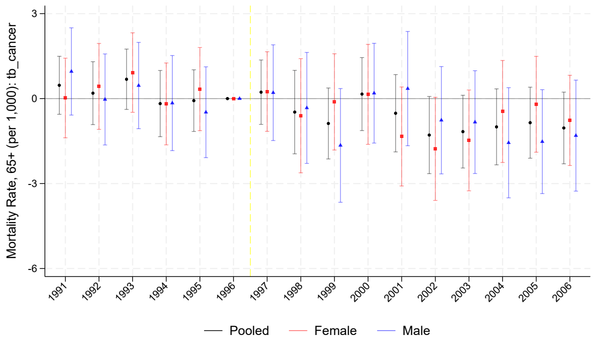


(c) Intensity in 1997 - Highly Marginalized Municipalities

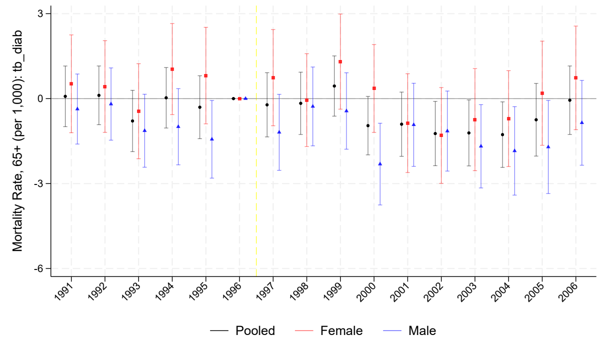
(d) Intensity in 1997 - All Municipalities

Notes: This figure documents geographic variation in Progresa enrollment for all municipalities, complementing Figure 1 in the main text which focuses on the highly marginalized subsample. Panel (a) and (b) map Progresa intensity across all municipalities for 1999 and 2005; panels (c) and (d) map intensity in 1997 separately for highly marginalized municipalities (the main sample) and all municipalities. Darker shading indicates higher program intensity.

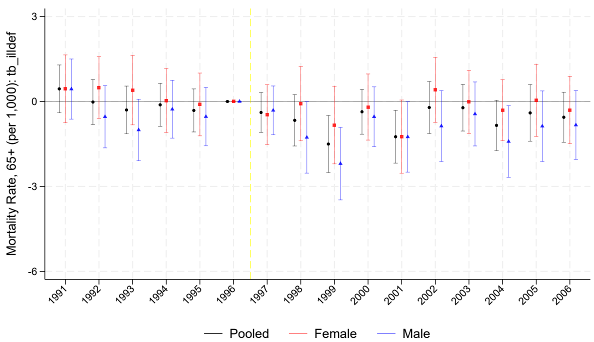
Figure A.3: Event Study: Effect of Progresa Enrollment Intensity on Cause-Specific Mortality



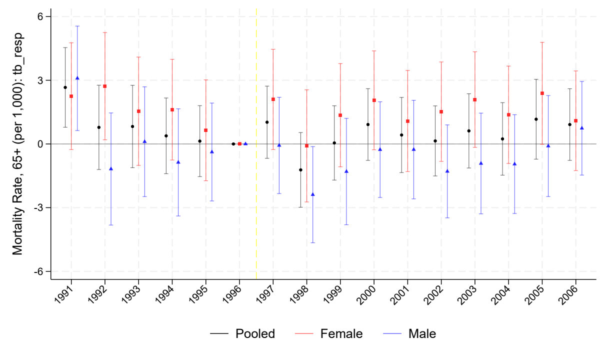
(a) Cancer



(b) Diabetes



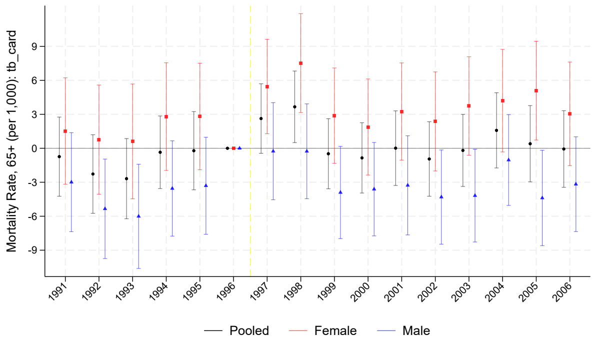
(c) Ill-Defined



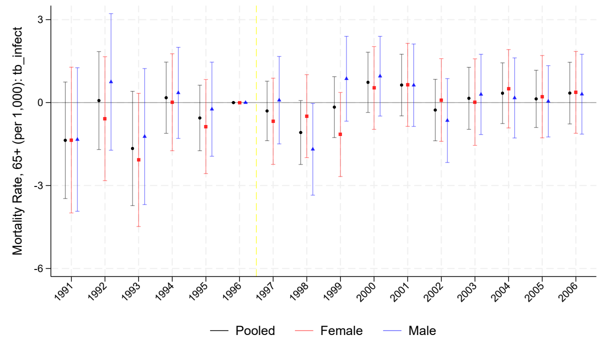
(d) Respiratory

Notes: Each panel displays interaction coefficients between Progresa intensity in 1999 and year indicators for the indicated cause of death. The outcome is the mortality rate for adults aged 65 and older per 1,000 inhabitants attributable to each cause. The specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level. The reference year is 1996. The sample is restricted to highly marginalized municipalities (*gm_mun_1990* = 4 or 5). Regressions are weighted by the population 65+. 95% confidence intervals shown.

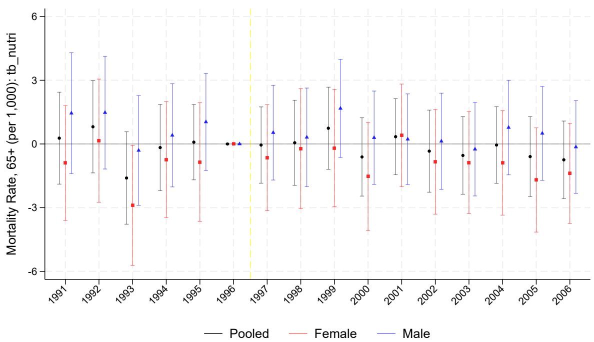
Figure A.4: Event Study: Effect of Progresa Enrollment Intensity on Cause-Specific Mortality



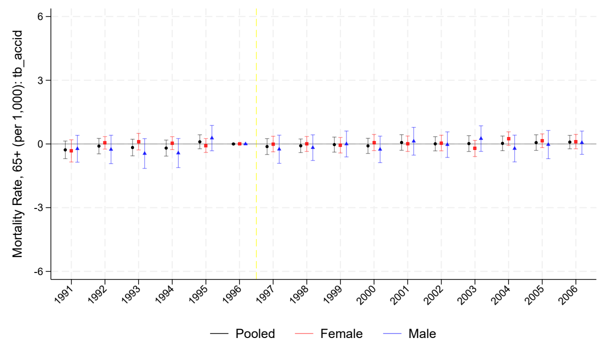
(a) Cardiovascular



(b) Infectious



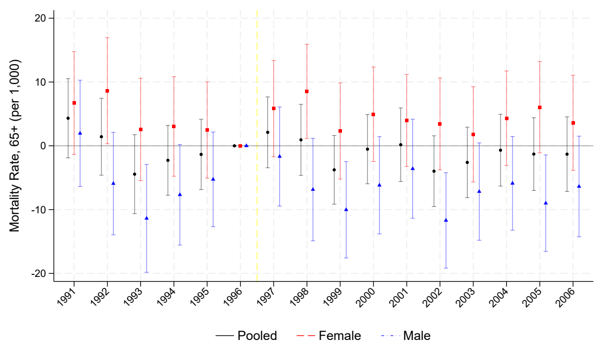
(c) Nutritional



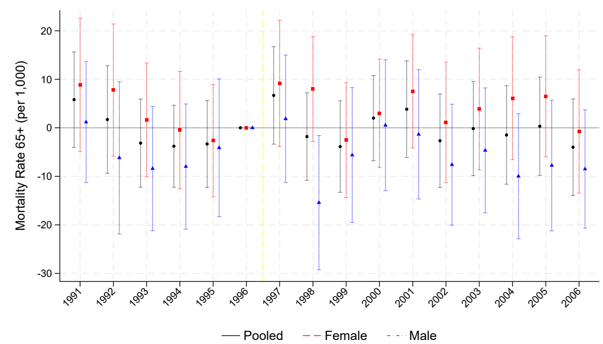
(d) Accidents

Notes: Each panel displays interaction coefficients between Progresa intensity in 1999 and year indicators for the indicated cause of death. The outcome is the mortality rate for adults aged 65 and older per 1,000 inhabitants attributable to each cause. The specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level. The reference year is 1996. The sample is restricted to highly marginalized municipalities (*gm_mun_1990* = 4 or 5). Regressions are weighted by population 65+. 95% confidence intervals shown.

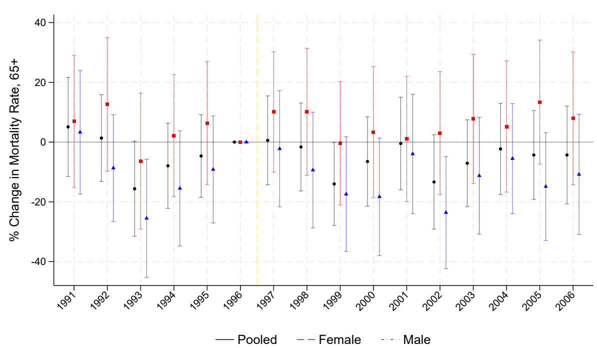
Figure A.5: Event Study: Effect of Progresa Enrollment Intensity on Mortality Rate by Sex and Alternative Specification



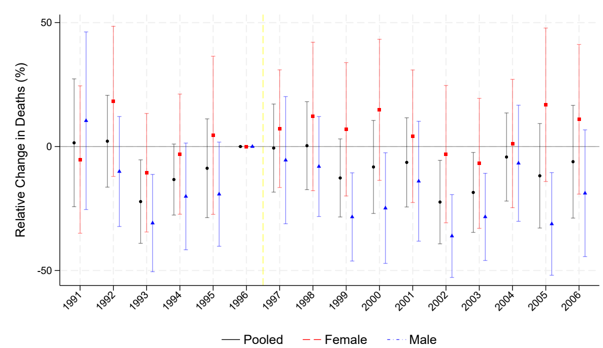
(a) Levels - Weighted



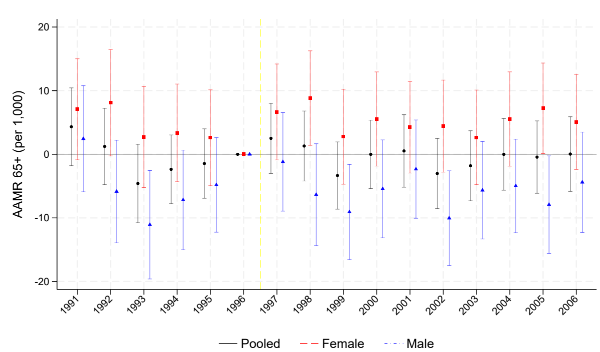
(b) Levels - Unweighted



(c) Logarithm



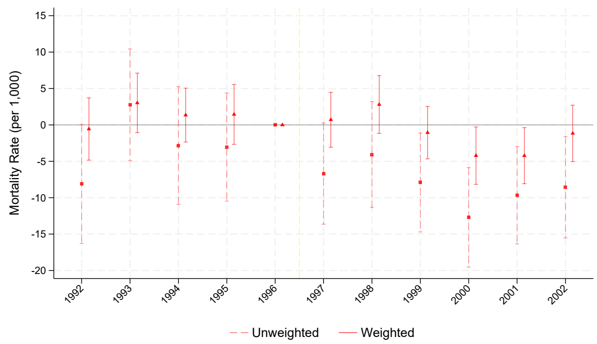
(d) Poisson



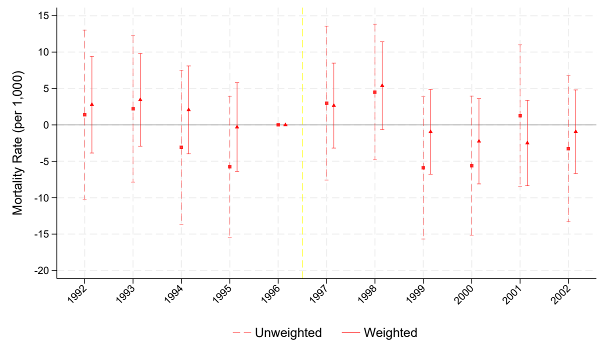
(e) AAMR Weighted +SP

Notes: All panels display interaction coefficients between Progresa intensity in 1999 and year indicators for pooled, female, and male populations, with 1996 as the reference year. The sample is restricted to highly marginalized municipalities (*gm_mun_1990* = 4 or 5). 95% confidence intervals shown. Panel (a) shows the baseline weighted specification using the mortality rate in levels, complementing the main Figure 2. Panel (b) shows the unweighted levels specification. Panel (c) uses the natural log of the mortality rate (municipality-years with zero deaths are excluded); coefficients are multiplied by 100 and express the approximate percentage change in the mortality rate per unit increase in Progresa intensity. Panel (d) uses Poisson pseudo-maximum likelihood with raw death counts as the outcome and population weights; coefficients are computed as $(\exp(\hat{\beta})-1)\times 100$ and express the percentage change in death counts per unit increase in intensity. Panel (e) uses the Age-Adjusted Mortality Rate (AAMR) in the weighted specification with Seguro Popular intensity controls, which directly standardizes for the age structure of each municipality's elderly population.

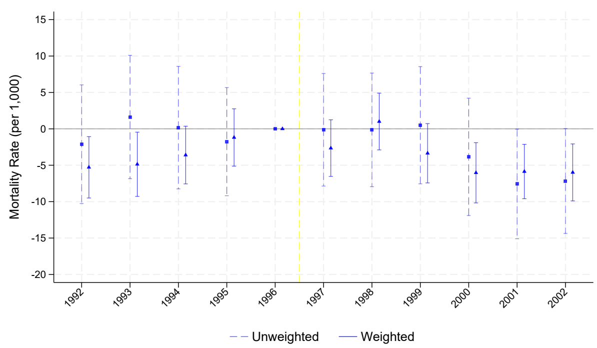
Figure A.6: Event Study: Effect of Progresa Enrollment Intensity on Mortality Rate by Sex and Sample, 1992–2002



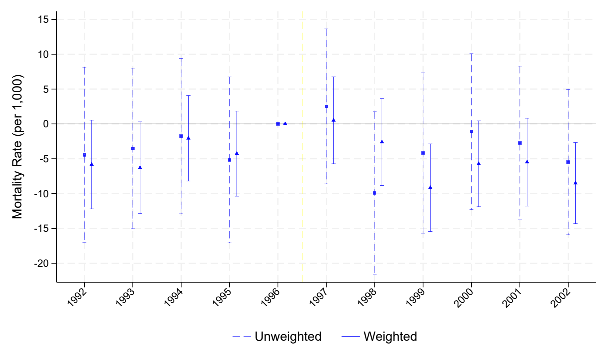
(a) Phase in 1998/1999 - Females



(b) Highly Marginalized - Females



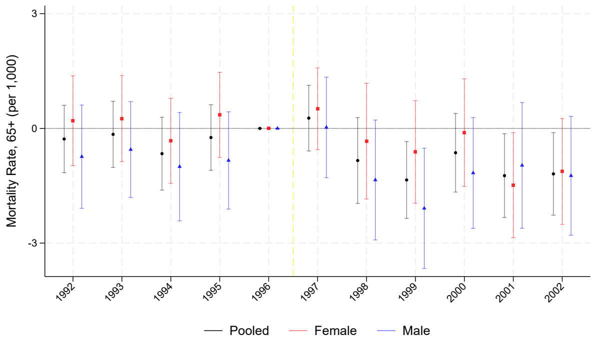
(c) Phase in 1998/1999 - Males



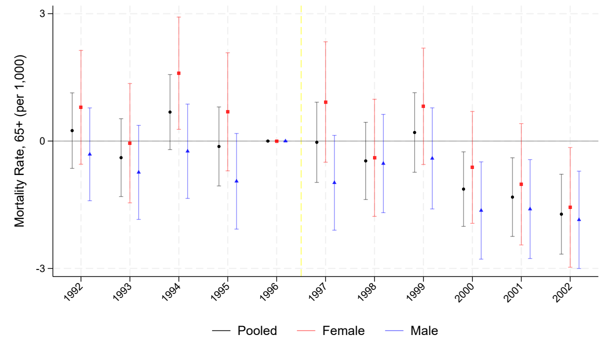
(d) Highly Marginalized - Males

Notes: This figure compares event study estimates across the two sample definitions used in this paper and in Barham and Rowberry (2013). Left panels (BR) restrict to municipalities incorporated into Progresa in 1998 or 1999; right panels (HM) restrict to highly marginalized municipalities (*gm_mun_1990* = 4 or 5). All regressions include municipality and year fixed effects; the reference year is 1996; the period is 1992–2002. 95% confidence intervals shown.

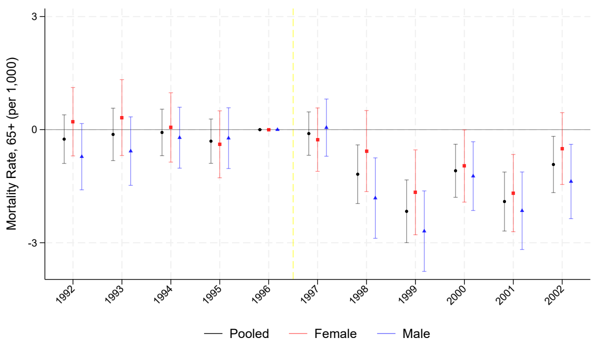
Figure A.7: Event Study: Effect of Progresa Enrollment Intensity on Cause-Specific Mortality, Highly Marginalized Sample, 1992–2002



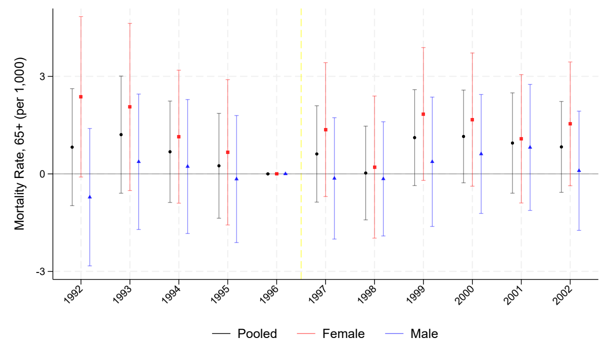
(a) Cancer



(b) Diabetes



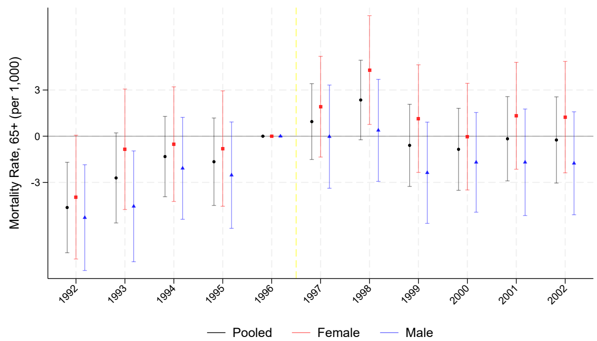
(c) Ill-Defined



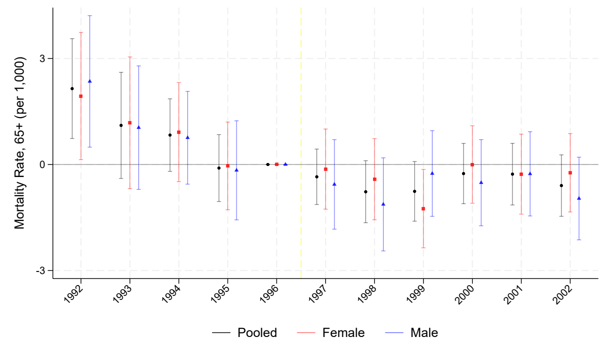
(d) Respiratory

Notes: Each panel displays interaction coefficients between Progresa intensity in 1999 and year indicators for the indicated cause of death. The outcome is the mortality rate for adults aged 65 and older per 1,000 inhabitants attributable to each cause. The specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level. The reference year is 1996. The sample is municipalities incorporated into Progresa in 1998 or 1999. Regressions are weighted. 95% confidence intervals shown.

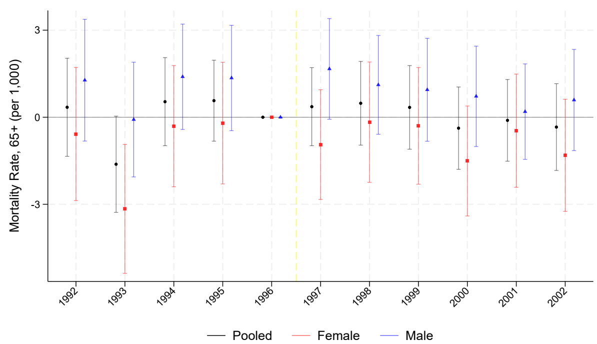
Figure A.8: Event Study: Effect of Progresa Enrollment Intensity on Cause-Specific Mortality, Highly Marginalized Sample, 1992–2002



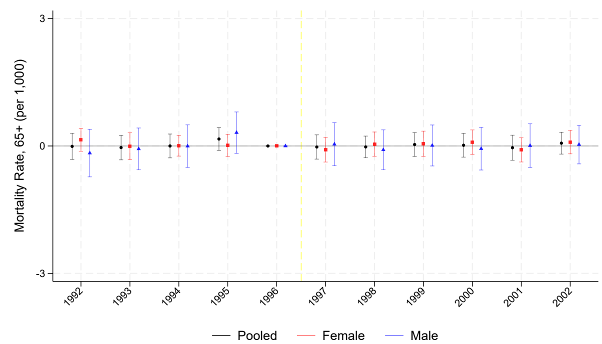
(a) Cardiovascular



(b) Infectious



(c) Nutritional



(d) Accidents

Notes: Each panel displays interaction coefficients between Progresa intensity in 1999 and year indicators for the indicated cause of death. The outcome is the mortality rate for adults aged 65 and older per 1,000 inhabitants attributable to each cause. The specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level. The reference year is 1996. The sample is municipalities incorporated into Progresa in 1998 or 1999. Regressions are weighted. 95% confidence intervals shown.

Table A.1: Effects of Progresa Enrollment Intensity on Cause-Specific Mortality Rate for Adults Aged 65 and Older

	Cancer	Diab.	IllDef	Resp.	Card.	Infect.	Nutri.	Accid.	Other
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Pooled									
Intensity 1999 x post	-0.887***	-0.508**	-0.574***	-0.313	1.530*	0.619	-0.102	0.102	-0.591
	(0.266)	(0.247)	(0.221)	(0.439)	(0.822)	(0.392)	(0.488)	(0.078)	(0.911)
RW p-value	[0.008]	[0.282]	[0.077]	[1.000]	[0.377]	[0.574]	[1.000]	[0.777]	[1.000]
Intensity 2005 x post	0.141	-1.340***	-0.621**	0.934	1.998**	-3.924***	-0.048	-0.059	0.109
	(0.304)	(0.298)	(0.244)	(0.599)	(1.013)	(0.501)	(0.559)	(0.086)	(1.010)
Mean (pre-1997)	1.70	1.63	1.00	4.34	14.01	3.68	4.35	0.21	15.54
Obs	17,904	17,904	17,904	17,904	17,904	17,904	17,904	17,904	17,904
Panel B: Females									
Intensity 1999 x post	-0.906***	-0.451	-0.487*	0.089	2.475**	0.837*	0.034	0.072	-1.043
	(0.321)	(0.336)	(0.270)	(0.561)	(0.989)	(0.470)	(0.580)	(0.077)	(1.099)
RW p-value	[0.044]	[0.902]	[0.498]	[1.000]	[0.100]	[0.498]	[1.000]	[1.000]	[1.000]
Intensity 2005 x post	0.343	-1.581***	-0.425	0.153	1.345	-4.336***	-0.455	-0.029	-0.656
	(0.359)	(0.417)	(0.287)	(0.721)	(1.206)	(0.548)	(0.665)	(0.087)	(1.211)
Mean (pre-1997)	1.61	2.04	1.11	4.17	14.51	3.64	4.68	0.10	14.26
Obs	17,904	17,904	17,904	17,904	17,904	17,904	17,904	17,904	17,904
Panel C: Males									
Intensity 1999 x post	-0.868**	-0.567*	-0.674**	-0.749	0.571	0.389	-0.273	0.129	-0.163
	(0.370)	(0.307)	(0.267)	(0.504)	(0.966)	(0.457)	(0.557)	(0.129)	(1.079)
RW p-value	[0.153]	[0.454]	[0.107]	[0.825]	[1.000]	[1.000]	[1.000]	[1.000]	[1.000]
Intensity 2005 x post	-0.070	-1.049***	-0.830***	1.736***	2.638**	-3.491***	0.360	-0.088	0.997
	(0.417)	(0.341)	(0.300)	(0.657)	(1.124)	(0.587)	(0.613)	(0.140)	(1.183)
Mean (pre-1997)	1.80	1.25	0.89	4.52	13.61	3.74	4.02	0.32	17.10
Obs	17,904	17,904	17,904	17,904	17,904	17,904	17,904	17,904	17,904

Notes: This table presents cause-specific difference-in-differences estimates for the main highly marginalized sample over 1991–2006. Each column corresponds to a specific cause of death. *Progresa* enrollment intensity is interacted with the post-1997 dummy following equation (1). Panel A reports pooled estimates; Panel B reports estimates for females; Panel C reports estimates for males. The sample is restricted to highly marginalized municipalities (*gm_mun_1990* = 4 or 5). All regressions include municipality and year fixed effects, control for Seguro Popular intensity, and are weighted by the population aged 65 and older. Standard errors are clustered at the municipality level. ***p< 0.01, **p< 0.05, *p< 0.1.

Table A.2: Functional Form and Outcome Robustness: Effect of Progresa Enrollment Intensity on Elderly Mortality

	Levels	Log	Poisson	AAMR
	(1)	(2)	(3)	(4)
Panel A: Pooled				
Intensity 1999 x post (1997-2006)	-0.777	-1.98	-3.06	0.057
	(1.464)	(4.20)	(5.76)	(1.493)
Intensity 2005 x post (1997-2006)	-2.787	-5.24	-12.85*	-3.401*
	(1.806)	(5.18)	(7.16)	(1.860)
Mean (1991-1996)	46.47	3.75	23.86	46.28
Panel B: Females				
Intensity 1999 x post (1997-2006)	0.592	2.52	5.51	1.395
	(1.824)	(5.48)	(7.37)	(1.850)
Intensity 2005 x post (1997-2006)	-5.637**	-17.27**	-22.27***	-6.448***
	(2.205)	(6.72)	(7.56)	(2.281)
Mean (1991-1996)	46.11	3.73	11.61	46.50
Panel C: Males				
Intensity 1999 x post (1997-2006)	-2.288	-2.87	-10.36*	-1.362
	(1.732)	(5.04)	(6.19)	(1.759)
Intensity 2005 x post (1997-2006)	0.214	-2.03	-2.37	-0.707
	(2.000)	(5.70)	(7.62)	(2.031)
Mean (1991-1996)	47.26	3.78	12.25	47.50
Obs	17,904	17,223	17,904	17,821
No. Mun	1,119	1,119	1,119	1,119
Seguro Popular	N	N	N	Y
Mean Intensity 1999 (%)	44.4	44.4	44.4	44.4

Notes: This table assesses robustness of the main DiD estimates to functional form and outcome definition. Column (1) reports the baseline specification using the mortality rate in levels (weighted). Column (2) uses the natural log of the mortality rate; reported coefficients are multiplied by 100 and express the approximate percentage change in the mortality rate per unit increase in Progresa intensity. Column (3) uses Poisson pseudo-maximum likelihood with raw death counts as the outcome and population weights; reported coefficients are computed as $(\exp(\hat{\beta}) - 1) \times 100$ and express the percentage change in death counts per unit increase in intensity. Column (4) replaces the mortality rate with the Age-Adjusted Mortality Rate (AAMR), which directly standardizes for the age structure of each municipality's elderly population, using the population-weighted specification with Seguro Popular intensity controls. All specifications include municipality and year fixed effects and are weighted by the population aged 65 and older. The sample is restricted to highly marginalized municipalities (*gm_mun_1990* = 4 or 5). Standard errors are clustered at the municipality level. ***p< 0.01, **p< 0.05, *p< 0.1.

Table A.3: Replication of Barham and Rowberry (2013): Effect of Progresa on Mortality Rate for Adults Aged 65 and Older

	Pooled	Females	Males
	(1)	(2)	(3)
Panel A: BR (2013)			
2-yr lagged Intensity	-6.37***	-6.46***	-6.42***
	(1.04)	(1.31)	(1.42)
Mean 1996	47.5	46.0	49.3
Obs	21,571	21,571	21,571
No. Mun	1,961	1,961	1,961
Panel B: Replication (Unweighted)			
2-yr lagged Intensity	-7.061***	-7.426***	-6.889***
	(1.256)	(1.559)	(1.711)
Panel C: Replication (Weighted)			
2-yr lagged Intensity	-3.740***	-4.210***	-3.322***
	(0.786)	(1.002)	(0.935)
Panel D: 1-yr Lag (Weighted)			
1-yr lagged Intensity	-3.457***	-4.255***	-2.681**
	(0.890)	(1.043)	(1.072)
Panel E: 3-yr Lag (Weighted)			
3-yr lagged Intensity	-2.071***	-1.830*	-2.357**
	(0.772)	(0.960)	(0.957)
Mean 1996	48.16	47.49	48.87
Obs	15,642	15,642	15,642
No. Mun	1,422	1,422	1,422
Mean Intensity 1999 (%)	31.6	31.6	31.6

Notes: This table replicates the main specification of Barham and Rowberry (2013) using the same sample (municipalities incorporated into Progresa in 1998 or 1999) and the same estimating equation (two-year lagged intensity, no post-interaction). The outcome is the mortality rate for adults aged 65 and older per 1,000 inhabitants. Row (A) reproduces their original unweighted estimate using a two-year lagged intensity variable and the BR sample (municipalities incorporated in 1998 or 1999); row (B) reports our close replication. Row (C) adds population weights, which attenuates the estimate by approximately 50% and reduces precision. Rows (D) and (E) replace the two-year lag with one-year and three-year lags, respectively. All regressions include municipality and year fixed effects. Standard errors are clustered at the municipality level and shown in parentheses. ***p< 0.01, **p< 0.05, *p< 0.1.

Table A.4: Sensitivity Analysis: Effect of Progresa Enrollment Intensity on Elderly Mortality, BR and High Marginalization Samples, 1992–2002

	BR Sample		*High Marginalization*	
	Unweighted	Weighted	Unweighted	Weighted
	(1)	(2)	(3)	(4)
Panel A: Pooled				
Intensity x Post (1997-2002)	-3.974***	-1.660**	-0.531	-1.884
	(1.214)	(0.807)	(2.122)	(1.287)
Mean (1991-1996)	47.45	47.45	46.65	46.65
Obs	15642	15642	13924	13924
No. Mun	1422	1422	1119	1119
Panel B: Females				
Intensity x Post (1997-2002)	-6.016***	-2.343**	-0.492	-2.175
	(1.599)	(1.008)	(2.490)	(1.536)
Mean (1991-1996)	46.69	46.69	46.45	46.45
Obs	15642	15642	13924	13924
No. Mun	1422	1422	1119	1119
Panel C: Males				
Intensity x Post (1997-2002)	-2.609	-1.002	-1.297	-1.686
	(1.681)	(1.032)	(2.569)	(1.589)
Mean (1991-1996)	48.49	48.49	47.27	47.27
Obs	15642	15642	13924	13924
No. Mun	1422	1422	1119	1119
Mean Intensity 1999 (%)	31.6	31.6	44.4	44.4

Notes: This table applies the interaction design of equation (4) to both the Barham-Rowberry sample and the full highly marginalized sample, using $\mathit{Intensity}_{1999} \times \mathit{Post}$ as the treatment variable over a 1992–2002 window. The outcome is the mortality rate for adults aged 65 and older per 1,000 inhabitants. Columns (1) and (2) report unweighted and population-weighted estimates for the Barham-Rowberry sample (municipalities incorporated into Progresa in 1998 or 1999); columns (3) and (4) report the corresponding estimates for the full highly marginalized sample (*gm_mun_1990* = 4 or 5). Panel A reports pooled estimates; Panel B reports estimates for females; Panel C reports estimates for males. All regressions include municipality and year fixed effects and Seguro Popular intensity controls. Standard errors are clustered at the municipality level. ***p< 0.01, **p< 0.05, *p< 0.1.

Table A.5: Effects of Progresa Enrollment Intensity on Cause-Specific Mortality Rate for Adults Aged 65 and Older, Barham-Rowberry Sample, 1992–2002

	Cancer	Diab.	IllDef	Resp.	Card.	Infect.	Nutri.	Accid.	Other
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Pooled									
Intensity 1999 x post	-0.816***	-0.924***	-0.614***	0.149	0.792*	-1.670***	-0.279	0.019	1.681***
	(0.188)	(0.183)	(0.134)	(0.281)	(0.470)	(0.210)	(0.248)	(0.048)	(0.498)
RW p-value	[0.000]	[0.000]	[0.000]	[1.000]	[0.370]	[0.000]	[0.784]	[1.000]	[0.004]
Mean (pre-1997)	2.48	2.58	0.91	4.86	15.84	2.56	3.54	0.27	14.39
Obs	15,642	15,642	15,642	15,642	15,642	15,642	15,642	15,642	15,642
Panel B: Females									
Intensity 1999 x post	-0.600***	-0.758***	-0.687***	-0.126	0.821	-1.792***	-0.447	-0.035	1.279**
	(0.215)	(0.271)	(0.170)	(0.350)	(0.577)	(0.227)	(0.308)	(0.043)	(0.579)
RW p-value	[0.037]	[0.037]	[0.000]	[0.845]	[0.591]	[0.000]	[0.591]	[0.845]	[0.137]
Mean (pre-1997)	2.13	3.15	0.94	4.43	16.35	2.46	3.86	0.12	13.25
Obs	15,642	15,642	15,642	15,642	15,642	15,642	15,642	15,642	15,642
Panel C: Males									
Intensity 1999 x post	-1.074***	-1.086***	-0.559***	0.450	0.702	-1.532***	-0.140	0.075	2.163***
	(0.262)	(0.212)	(0.159)	(0.334)	(0.565)	(0.261)	(0.263)	(0.083)	(0.641)
RW p-value	[0.000]	[0.000]	[0.003]	[0.714]	[0.714]	[0.000]	[0.734]	[0.734]	[0.004]
Mean (pre-1997)	2.85	2.04	0.89	5.28	15.38	2.69	3.21	0.42	15.73
Obs	15,642	15,642	15,642	15,642	15,642	15,642	15,642	15,642	15,642

Notes: This table replicates the cause-specific analysis of Appendix Table A.1 using the Barham-Rowberry sample (municipalities incorporated into *Progresa* in 1998 or 1999) over 1992–2002. Each column corresponds to a specific cause of death. *Progresa* enrollment intensity ($Intensity_{1999}$) is interacted with the post-1997 dummy following equation (4). Panel A reports pooled estimates; Panel B reports estimates for females; Panel C reports estimates for males. All regressions include municipality and year fixed effects, control for Seguro Popular intensity, and are weighted by the population aged 65 and older. Standard errors are clustered at the municipality level. ***p< 0.01, **p< 0.05, *p< 0.1.